Project Insight: Fine-Grained Rehabilitation Error Understanding

Proposed direction: Error Type + Body Region Localization using KERAAL skeleton data

1. The Main Idea

We want to go beyond simply asking whether an AI model can classify a rehabilitation exercise error. The proposed project investigates whether skeleton-based movement data can identify both **what error occurred** and **where in the body the error is associated**.

> **Simple idea:**
> Movement → AI → Error type + Body region

For example, instead of producing only “Error A,” the system could produce an error category together with the relevant body-region category, if the KERAAL annotations support reliable labels for that sample.

2. Why We Need a Different Research Question

The KERAAL literature already includes skeleton-based rehabilitation error classification. Therefore, simply using another model for the same classification task would not by itself make a strong research contribution.

Our focus is **fine-grained error understanding**: whether movement characteristics can help identify the body region associated with an error, and whether this information can improve or explain error recognition.

> **The contribution should be the research question and analysis, not just the choice of LSTM, TCN, or another architecture.**

3. What Data We Need

The main dataset is KERAAL. The project proposal describes skeleton sequences, predefined error labels, subject-independent evaluation, and kinematic representations such as joint angles, joint velocity, and left-right symmetry.

> Before building the final model, we must inspect the actual KERAAL annotation files. We need to verify the exact body-region labels, sample counts, how labels match skeleton sequences, and whether the annotations are consistent enough for supervised learning.

4. Complete Project Process

Step	Stage	What we do
1	Get KERAAL	Access the skeleton sequences and annotation files.
2	Understand annotations	Identify exercise, error, participant, and available body-region information.
3	Build dataset index	Connect each skeleton sequence to its participant and labels.
4	Inspect skeleton	Check frames, joints, coordinates, missing values, and sequence lengths.
5	Visualize movement	Plot/play skeleton sequences so the team understands the movement.
6	Preprocess	Root-center and normalize the skeleton according to the planned protocol.
7	Create features	Compare raw coordinates with joint angles, velocity, and left-right symmetry.
8	Build baseline	Train a basic error-classification model first.
9	Add body-region task	Predict error type and body region together if annotations support it.
10	Subject-independent test	Keep test participants completely separate from training participants.
11	Analyze results	Compare representations, error classes, body regions, and model behavior.

Step	Stage	What we do
12	Demo + paper	Present the pipeline and write the findings and limitations.

5. What the AI Actually Learns

Each rehabilitation repetition is a sequence of skeleton frames. The model receives movement information over time rather than one static image.

Raw representation: X, Y, Z coordinates of joints over time.

Kinematic representation: movement descriptions such as joint angles, joint velocity, and left-right symmetry.

We compare these representations to investigate whether more meaningful movement features help generalization to participants the model has not seen before.

6. Proposed Model

Skeleton sequence
↓
Preprocessing / normalization
↓
Raw coordinates OR kinematic features
↓
Temporal model
↓
Error type + **Body region**

If body-region annotations are too incomplete or inconsistent, we should not force this second prediction task. The data inspection determines whether this direction is feasible.

7. Experiments

Experiment 1 — Baseline: basic error classification on subject-disjoint data.

Experiment 2 — Raw vs kinematic: compare raw coordinates with joint angles, velocity, symmetry, and combinations.

Experiment 3 — Body region: evaluate whether movement can predict the annotated body-region category.

Experiment 4 — Multi-task: compare error-only prediction with joint error + body-region prediction.

Experiment 5 — Error analysis: determine which errors and regions are difficult and investigate possible movement-feature explanations.

8. The Actual Insight

The paper should not only say “our model achieved X%.” The important result is what the experiments teach us.

> **Possible insight to investigate:** different rehabilitation errors may depend on different movement characteristics. Some may be easier to recognize from joint-angle changes, while others may depend more on velocity or left-right asymmetry. We can investigate whether these characteristics also correspond to the body region associated with the error.

This is a hypothesis, not a result we can claim beforehand. The final paper must report what the experiments actually show.

9. What Makes This Different

Weak question	Stronger research question
Can LSTM classify KERAAL errors?	What movement representations help error recognition generalize to unseen participants?
Is TCN better than LSTM?	Which movement characteristics are associated with different error categories?

Can we get high accuracy?	Can the model provide finer-grained information about the body region associated with an error?
Which model wins?	What do model errors reveal about difficult rehabilitation movements?

10. Evaluation

The main evaluation should remain subject-independent: participants in the test set must not appear in the training set. The proposal identifies Macro-F1 as the main metric, supported by balanced accuracy, per-class recall, and confusion matrices.

11. Final Demo

A simple demonstration could show: Skeleton sequence → AI model → Error category + associated body-region category. Any feedback should describe observable movement only and should not claim to diagnose a condition or prescribe treatment.

12. Team Roles

Area	Main work
AI / Data Science	Dataset processing, feature extraction, model training, evaluation, visualization.
Biology / Clinical interpretation	Understand rehabilitation movement and error definitions; interpret body-region and feature findings; review feedback wording.
Mechanical / Control	Skeleton geometry, joint relationships, movement dynamics, kinematic feature design.
Project support	Data organization, experiment records, figures, documentation, demo preparation.

13. What We Do First

Do not start with the neural network. First establish data feasibility:

• Get KERAAL and the annotation files.

• Identify every available label.

• Check body-region sample counts and consistency.

• Match labels to skeleton sequences.

• Visualize several examples.

• Then finalize the model and experiments.

14. One-Sentence Summary

> “We want to move beyond simply detecting what rehabilitation error happened and investigate whether skeleton kinematics can identify both the error and the body region associated with it, while studying which movement features help this generalize to unseen participants.”

Feasibility note: This is a proposed research direction, not a guaranteed publishable result. Its feasibility depends on the actual KERAAL annotations and what the experiments reveal.